

$$\begin{aligned}
 1) \quad 1 &= \int_0^1 \int_0^{1-x} (cx+1) dy dx \\
 &= \int_0^1 (cx+1)(1-x) dx = \int_0^1 cx + 1 - cx^2 - x dx \\
 &= c \cdot \frac{1}{2} + 1 - c \cdot \frac{1}{3} - \frac{1}{2} = c \cdot \frac{1}{6} + \frac{1}{2} \\
 &\Rightarrow \boxed{c = 3}.
 \end{aligned}$$

$$\begin{aligned}
 EXY &= \int_0^1 \int_0^{1-x} (xy)(cx+1) dy dx \\
 &= \int_0^1 x(cx+1) \cdot \frac{(1-x)^2}{2} dx \\
 &= \frac{1}{2}c \int_0^1 x^2(1-x)^2 dx + \frac{1}{2} \int_0^1 x(1-x)^2 dx \\
 &= \frac{1}{2}c \left(\frac{1}{3} - \frac{2}{4} + \frac{1}{5} \right) + \frac{1}{2} \left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4} \right) \\
 &= \frac{c}{60} + \frac{1}{24} = \boxed{\frac{11}{120}}
 \end{aligned}$$

$$\begin{aligned}
 \exists X: \text{for } x \in [0,1]: f_X(x) &= \int_0^{1-x} (cx+1) dy \\
 &= (cx+1)(1-x)
 \end{aligned}$$

$$\Rightarrow EX = \int_0^1 x(cx+1)(1-x) dx$$

$$= \int_0^1 (cx^2 + x)(1-x) dx$$

$$= \int_0^1 cx^2 + x - cx^3 - x^2 dx$$

$$= c \cdot \frac{1}{3} + \frac{1}{2} - \frac{c}{4} - \frac{1}{3}$$

$$= \frac{c}{12} + \frac{1}{6} = \boxed{\frac{5}{12}}$$

Аналогично $EX^2 = \int_0^1 cx^3 + x^2 - cx^4 - x^3 dx$

$$= c \left(\frac{1}{4} - \frac{1}{5} \right) + \frac{1}{3} - \frac{1}{4}$$

$$= c \cdot \frac{1}{20} + \frac{1}{12} = \boxed{\frac{7}{30}}$$

$$\Rightarrow DX = \frac{7}{30} - \left(\frac{5}{12} \right)^2 = \frac{43}{720} \approx 0.0597$$

За γ : за $y \in [0, 1]$: $f_\gamma(y) = \int_0^y cx + 1 dx$

$$= \frac{c}{2}(1-y)^2 + 1-y$$

$$\Rightarrow EY = \frac{C}{2} \int_0^1 y(1-y)^2 dy + \int_0^1 y(1-y) dy$$

$$= \frac{C}{2} \left[\frac{1}{2} - \frac{2}{3} + \frac{1}{4} \right] + \frac{1}{2} - \frac{1}{3}$$

$$= \frac{C}{24} + \frac{1}{6} = \boxed{\frac{7}{24}}$$

Answer: $EY^2 = \frac{C}{2} \int_0^1 y^2(1-y)^2 dy + \int_0^1 y^2(1-y) dy$

$$= \frac{C}{2} \left[\frac{1}{3} - \frac{2}{4} + \frac{1}{5} \right] + \frac{1}{3} - \frac{1}{4}$$

$$= \frac{C}{60} + \frac{1}{12} = \boxed{\frac{2}{15}}$$

и тогда $DY = 2/15 - (7/24)^2$

$$= 139/2880 \approx 0.0483.$$

Следовательно,

$$\text{Cor}(X, Y) = \frac{EXY - EXEY}{\sqrt{DX \cdot DY}} = \frac{\frac{11}{120} - \frac{5}{12} \cdot \frac{7}{24}}{\sqrt{\frac{43}{720} \cdot \frac{139}{2880}}} \approx \underline{\underline{0.5562}}$$

$$\star \text{Cor}(X, Y) = \frac{43}{1440} \approx -0.030.$$

$$2. f_{X|Y}(x|1/2) = \frac{f_{X,Y}(x, 1/2)}{f_Y(1/2)}$$

$$= \frac{cx+1}{7/8} \quad \text{for } x \in [0, 1/2]$$

$$\Rightarrow E[X|Y=1/2] = \int_0^{1/2} x \cdot \frac{cx+1}{7/8} dx$$

$$= \frac{8}{7} \left[c \cdot \frac{1}{24} + \frac{1}{8} \right] = \boxed{\frac{2}{7}} \quad \text{20-2857.}$$

$$2) 1. P(N(\mu, 10^2) < 250) = 15\%$$

$$P(N(0,1) < \frac{250 - \mu}{10^2}) = 15\%$$

От таблицы: $\frac{250 - \mu}{10^2} \approx -1,04$

$$\Rightarrow \mu \approx 250 + 100 \cdot 1,04$$

$$= \boxed{260,4}$$

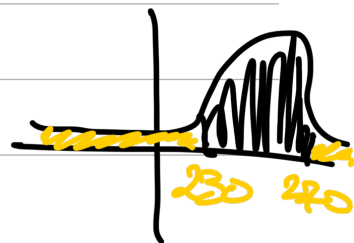
$$2. P(N(\mu, 10^2) > 280)$$

$$= P(N(0,1) > \frac{280 - \mu}{10})$$

$$\approx 1 - \Phi(1,96) = \Phi(-1,96)$$

$$\approx 0.025 = \boxed{2,5\%}$$

$$3. P(230 \leq N(250, \sigma^2) \leq 270)$$



$$= 1 - 2P(N(250, \sigma^2) < 230)$$

$$= 1 - 2P(N(0, 1) < \frac{230 - 250}{\sigma})$$

по условию 95%

$$\Rightarrow \text{Ищем } \sigma, \text{ т.е. } \Phi\left(-\frac{20}{\sigma}\right) = 0,015$$

от табл.

$$\Rightarrow -\frac{20}{\sigma} \approx -2,17 \Rightarrow \boxed{\sigma \approx 9,2166}$$

3) 1. Нека X_i = резултат на ст. i на заг. 1.
 $\sim 10 \cdot \text{Ber}(\frac{8}{10})$

$$EX_i = 10 \cdot \frac{8}{10} = 8 \quad \text{и} \quad DX_i = 100 \cdot \frac{8}{10} \cdot \frac{2}{10} = 16$$

Нека $S_n = X_1 + \dots + X_n$. От ЛТТ: $\frac{S_n - n \cdot 8}{4\sqrt{n}} \xrightarrow{d} N(0,1)$

$$P\left(\frac{S_{100}}{100} < 8\right) = P(S_{100} < 800) = P\left(\frac{S_{100} - 100 \cdot 8}{4\sqrt{100}} < 0\right)$$

ЛТТ, $100 > 30$
 $\approx P(N(0,1) < 0) = \boxed{\frac{1}{2}}$

Може и да се разглежда по симетрия, т.к.
 $S_{100} \sim \frac{\text{Bin}(100, 8/10) \cdot 10}{100}$

2. Нека Y_i = # точки на стъг. i .
 $\sim X_i^{(1)} + X_i^{(2)} + X_i^{(3)} + X_i^{(4)}$, където
 $X_i^{(1)} \sim 10 \cdot \text{Ber}(\frac{8}{10})$, $X_i^{(2)} \sim 10 \cdot \text{Ber}(\frac{7}{10})$, $X_i^{(3)} \sim 10 \cdot \text{Ber}(\frac{6}{10})$, $X_i^{(4)} \sim 10 \cdot \text{Ber}(\frac{4}{10})$,
 са нез. сл. вел., които кореспондират с точките на i
 по съответните застави.

Да отбележим, че $EX_i = 10\left(\frac{8}{10} + \frac{7}{10} + \frac{6}{10} + \frac{4}{10}\right) = 25$
 и $DY_i = 100\left(\frac{8 \cdot 2 + 7 \cdot 3 + 6 \cdot 4 + 4 \cdot 6}{100}\right) = 85$

Означается же i и $Z_i := 2 + \frac{Y_i}{40}$

$$\Rightarrow E Z_i = E \left[2 + \frac{Y_i}{40} \right] = 4,5$$

$$D Z_i = D \left[2 + \frac{Y_i}{40} \right] = \frac{D Y_i}{1600} = \frac{17}{2000}$$

3) От $\sqrt{n}$: $\frac{\sum_{i=1}^n Z_i - 4,5}{\sqrt{17/2000} / \sqrt{n}} \xrightarrow[n \rightarrow \infty]{d} N(0,1)$

Следовательно $P \left(\frac{Z_1 + \dots + Z_{100} - 4,5}{100} > \frac{1}{100} \cdot 4,5 \right)$

$$= P \left(\frac{\sum_{i=1}^{100} Z_i - 4,5}{\sqrt{17/2000}} > \frac{4,5/100}{\sqrt{17/2000}} \right)$$

$\sqrt{n}, 100 > 30$
 $\approx P(N(0,1) > 0,49) \approx \underline{\underline{31,21\%}}$

4.1. Here $X \sim \text{Exp}(1/100)$. Требуется отг. е:

$$P(X < 200) + P(X \in [300, 400])$$

$$= 1 - P(X \geq 200) + P(X \geq 300) - P(X \geq 400)$$

$$= 1 - e^{-2} + e^{-3} - e^{-4} := q \approx 89,61 \% , \text{ т.к.}$$

$$P(X > t) = e^{-\lambda t}$$

2. У нас 2 игрока 200 руб. бер $P(X > 300) = e^{-3}$

Следов. ожидаемая чистая прибыль е

$$e^{-3} \cdot (x - 200) + (1 - e^{-3})x$$

$$= x - 200/e^3 . \text{ Требуем } x:$$

$$x - 200/e^3 = 10 \Rightarrow x \approx 19,96 \text{ руб.}$$

3. У нас 1 игрок $x = 10 + 200/e^3$ и у нас 2

$$\text{и покрывает } \begin{cases} X, & X < 300 \\ X - 200, & X \geq 300 \end{cases}, \text{ т.к.}$$

$$\text{средно } EX - P(X \geq 300) \cdot 200$$

$$= 100 - e^{-3} \cdot 200 \approx 90,04 \text{ KB.}$$

В такъв случай, ако ухвата на разликата е y , тогава y , т.е.

$$y - [\text{загуба}] = 10$$

$$y - \left(x + 100 - \frac{200}{e^3} \right) = 10$$

$$\underline{y = 120}$$

(Можем ли да кажем, че отг. е 120 KB. от р-ето на разликата на цена?!)

Ако няма презастраховка

$$y - [\text{загуба}] = 10 \Rightarrow \underline{y = 110}.$$